MD EMON PRODHAN

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ACADEMIC CREDENTIALS

Master of Science (M.S.) in Statistics **May 2025 - Expected Dec 2026**

Shahjalal University of Science and Technology, Sylhet-3114, Bangladesh

- **Current CGPA: 3.76 / 4.00 (After 2nd of 3 Semesters)**
- ***M.S. Thesis:*** Gut health and its awareness levels among university students in Sylhet [Supervisor: **Dr. Mohammad Ohid Ullah**]

Bachelor of Science (B.Sc.) in Statistics **Jan 2020 - May 2025**

Shahjalal University of Science and Technology, Sylhet-3114, Bangladesh

- **CGPA: 3.84 / 4.00 with Distinction**
- ***Undergraduate Project:*** Determinants of early childhood mortality in Bangladesh [Supervisor: **Dr. Md Zakir Hossain**]

PUBLICATIONS AND PRESENTATIONS

Peer-Reviewed Journal Articles

Prodhan, M.E., Alam, A., Subha, S. et al. Learning environment at home and primary school dropout status in Bangladesh: an analysis of multiple indicator cluster survey. ***PLOS ONE***. 2026. **https://doi.org/10.1371/journal.pone.0360931**

Alam, A., **Prodhan, M.E.**, Akhter, T. et al. SWPER-Measured women's empowerment and mental health in Bangladesh: a cross-sectional study from 2022 demographic and health survey. ***BMC Women's Health*** 26, 119 (2026). **https://doi.org/10.1186/s12905-026-04300-w** [***Joint First Author***]

Alam, A., **Prodhan, M.E.**, Islam, F. et al. The effects of household water, sanitation, and hygiene on early childhood education enrollment: evidence from the 2022 Bangladesh demographic and health survey. ***BMC Public Health*** 25, 2732 (2025). **https://doi.org/10.1186/s12889-025-23968-7** [***Joint First Author***]

Conference Proceedings

Alam, A., **Prodhan, M.E.**, Hasan, M.N. et al. Deep learning–based multivariate time series forecasting of acute respiratory infections among Rohingya refugees in Cox's Bazar, Bangladesh. In: ***2025 28th International Conference on Computer and Information Technology (ICCIT)***; 2025; Cox's Bazar, Bangladesh. 2025. p. 3847-52. **https://doi.org/10.1109/ICCIT68739.2025.11491162** [***Joint First Author***]

Prodhan, M.E., Alam, A., Hasan, M.N. et al. A comparative study of machine learning algorithms in classifying and predicting exclusive breastfeeding infants in Bangladesh. In: ***2024 27th International Conference on Computer and Information Technology (ICCIT)***; 2024 Dec 19-21; Cox's Bazar, Bangladesh. 2024. p. 2789-93. **https://doi.org/10.1109/ICCIT64611.2024.11022563**

Hasan, M.N., **Prodhan, M.E.**, Alam, A. et al. A comparative study of machine learning algorithms in predicting acute respiratory infection of under-five children in Bangladesh using imbalanced data. In: ***2024 IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON)***; 2024; Dhaka, Bangladesh. 2024. p. 274-8. **https://doi.org/10.1109/BECITHCON64160.2024.10962602**

Oral/Poster Presentations

Prodhan, M.E., Miah, M.S., Ullah, M.O. Impact of pesticide use on farmers' health in north-eastern Bangladesh. Oral presentation at: ***International Conference on Emerging Frontiers in Advanced Sciences and Technologies (EFAST 2026)***; June 27–28, 2026; organized by Pabna University of Science and Technology (PUST), Bangladesh, and Universiti Malaysia Perlis (UniMAP), Malaysia.

Prodhan, M.E., Altaf, M.A.I., Pranta, J.P. Raising healthy children: The role of maternal education and mass media and ICT access in preventing child malnutrition. Poster presented at: ***International Seminar and StatFest 2023***; March 11–12 2023; organized by Department of Statistics, Shahjalal University of Science and Technology (SUST), Bangladesh.

Rakib, I.K., **Prodhan, M.E.**, Rahman, F., et al. Determinants and prevalence of acute respiratory infections, diarrheal diseases, and febrile illnesses among under-five children in Bangladesh: Analysis of the 2022 BDHS. Poster presented at: ***International Conference on Applied Statistics and Data Science (ICASDS 2025)***; December 28–29, 2025; organized by the Institute of Statistical Research and Training (ISRT), University of Dhaka, Bangladesh.

Manuscripts Under Review

Prodhan, M.E., Rakib, I.K. Determinants of high-risk fertility behavior among reproductive-age women in Bangladesh: A multilevel analysis of the 2022 Demographic and Health Survey. Under review at ***Health Science Reports***.

Masum, A.A., **Prodhan, M.E.**, Muhtasim, K.R. et al. Urban-rural disparities in non-food and health expenditures across income groups in Bangladesh: Evidence from the Household Income and Expenditure Survey 2022. Under review at ***BMC Public Health***.

RESEARCH EXPERIENCE

Research Assistant **Oct 2025 - Present**

Shahjalal University of Science and Technology, Sylhet-3114, Bangladesh

Project: *Impact of pesticides use on farmers' health in north-eastern part of Bangladesh*
Principal Investigator: **Dr. Mohammad Ohid Ullah**
Funding: Research Grant, SUST Research Centre
Role: Research Assistant

- Conducted literature reviews and synthesized findings for public health and social science research projects.
- Assisted in preparing manuscripts, reports, and conference materials for publication and dissemination.
- Performed statistical analysis and interpretation using R, SPSS, and STATA.
- Supported data visualization and presentation of research findings for academic and public audiences.

HONORS AND RECOGNITIONS

- **Master's Thesis Grants**
  Awarded by Research Centre, Shahjalal University of Science and Technology.
- **2nd Runner-up, 13th National Undergraduate Mathematics Olympiad 2022 (Sylhet Region)**
  Bangladesh Mathematical Society – AF Mujibur Rahman Foundation, November 2022
- **Best Paper Presentation**
  International Conference on Emerging Frontiers in Advanced Sciences and Technologies (EFAST 2026)
- **Finalist, Robi Datathon 3.0**
  Robi Axiata Limited, May 2024

- Awarded the University Scholarship multiple times for outstanding academic performance during the Bachelor of Science (B.Sc.) program.

SOFTWARE EFFICIENCY

- ***Programming*:** R, Python
- ***Data Analysis*:** SPSS, STATA, SAS, ArcGIS
- ***Productivity & Typesetting***: Microsoft Office, LaTeX

WORKSHOP/ CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI & Stanford University (September 2024; Credential ID: **Q7671CN2RPMG**)
- Data Analysis with R Programming — Google (April 2024; Credential ID: **W6AC7ZA2XE8F**)
- Data Science: R Programming Complete Diploma — Udemy (March 2024; Credential ID: **UC-df5fbbc5-f3a9-44b0-ab0f-42b0d8b1899b**)
- Introduction to Discrete Mathematics for Computer Science Specialization — University of California San Diego (July 2020; Credential ID: **RZUKRBLCJZL7**)
- Fundamentals of Research Methodology — SUST Research Centre (May 2025)

EXTRACURRICULAR EXPERIENCES

- **Peer Reviewer, BMJ Public Health & Elsevier Intelligence-Based Medicine**
  Contributed to the peer-review process for interdisciplinary research spanning public health, epidemiology, biomedical research, artificial intelligence, and evidence-based medicine.
- **Lead Instructor, SUST Data Science Club**
  Conducted training sessions on data analysis and basic statistical tools
- **Global Burden of Disease (GBD) Collaborator, Institute for Health Metrics and Evaluation (IHME)**
  Participate in data review, technical consultations, webinars, and collaborative research supporting the Global Burden of Disease Study